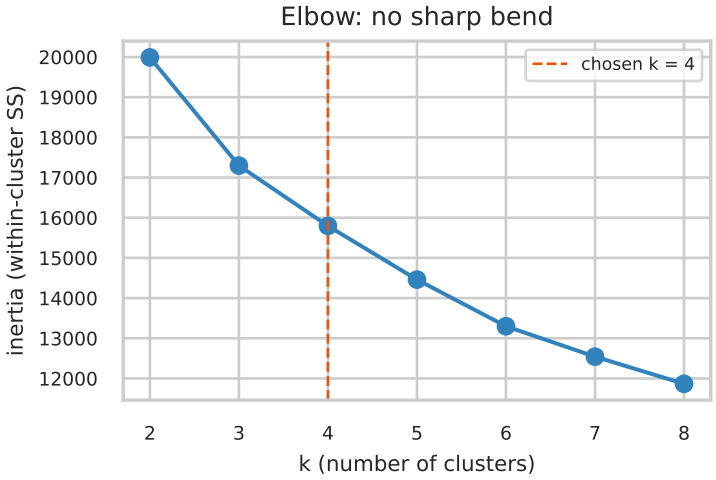
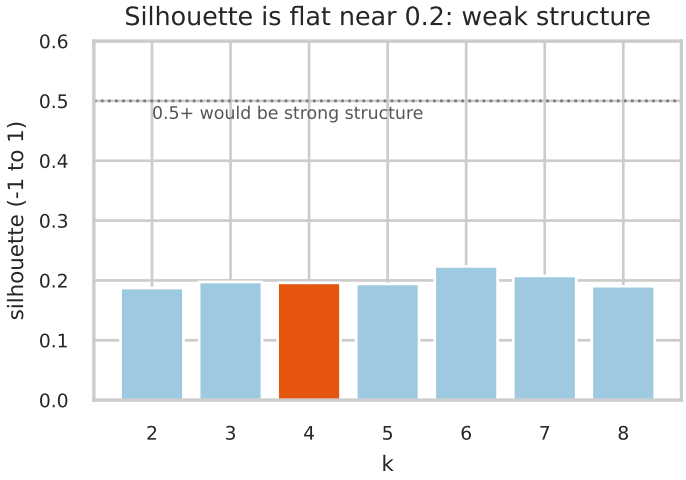
Day 19 Cheat Sheet: k-Means Clustering

Week 3: unsupervised patient segments, scaling, choosing k, profiling, stability

TERM	WHAT IT MEANS	CODE
Unsupervised	no label is used to fit; find structure in X only	KMeans().fit(X)
k-means	assign to nearest center, move centers, repeat	KMeans(n_clusters=k)
Scaling	distances need comparable units (z-scores)	StandardScaler()
Inertia	sum of squared distances to own center (lower=tighter)	km.inertia_
Elbow	k where inertia stops dropping fast	loop k = 2..8
Silhouette	-1 to 1; near 0 = overlapping, 0.5+ = clear groups	silhouette_score
n_init	restarts from different seeds; keeps the best	n_init=10
Profile	segment means in real units; give each a plain name	groupby().mean()
PCA (for plots)	squash 6 columns to 2 axes just to draw them	PCA(n_components=2)
Stability (ARI)	1.0 = same groups across seeds; check before you share	adjusted_rand_score





THIS DATA (same simulated days 14-18 discharges, seed 14): 4,000 patients, 6 features, readmit30 left OUT of the fit
Unscaled k = 4: clusters split almost only on age (mean ages 47, 60, 71, 84) because age has the widest spread
Scaled: silhouette 0.19 to 0.22 for k = 2..8, no sharp elbow. We pick k = 4 because the groups are easy to name
Segments: 0 no diabetes + follow-up visit (1286), 1 diabetes (1200), 2 no diabetes + no follow-up (1092), 3 long stay ~12 days (422)
Readmission AFTER fitting: 6.8%, 10.1%, 9.1%, 12.3% vs overall 9.0%. Stability across 3 other seeds: ARI 1.00

HABITS THAT KEEP YOU HONEST

- Always scale first; otherwise the column with the biggest numbers decides the clusters
- k-means always returns k groups, even in data with no real groups; check silhouette
- Choose k with metrics AND usefulness: can a care team name and act on each segment?
- Fit on features only; bring the outcome in afterwards to describe the segments
- Binary flags (diabetes, follow-up) dominate scaled distances; say so when you present

```
Xs = StandardScaler().fit_transform(df[features])
km = KMeans(n_clusters=4, n_init=10, random_state=19).fit(Xs)
df["segment"] = km.labels_
df.groupby("segment")[features + ["readmit30"]].mean()
```

Infographic: Finding Patient Segments Without a Label

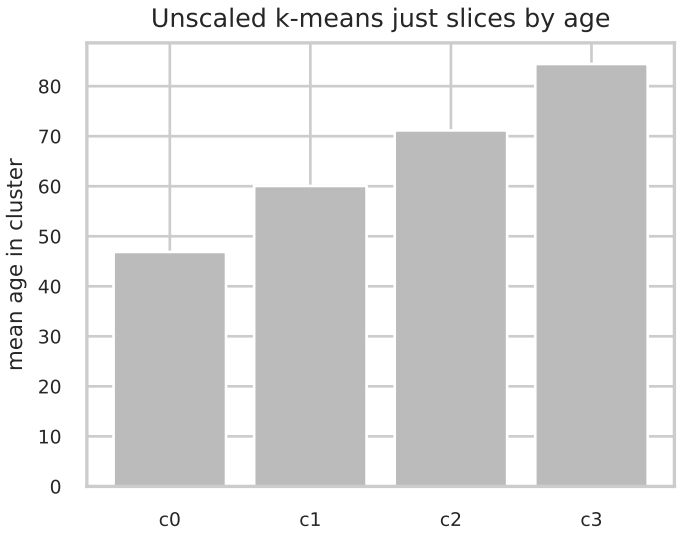
Same simulated readmissions data. k-means groups similar patients, then we look at outcomes

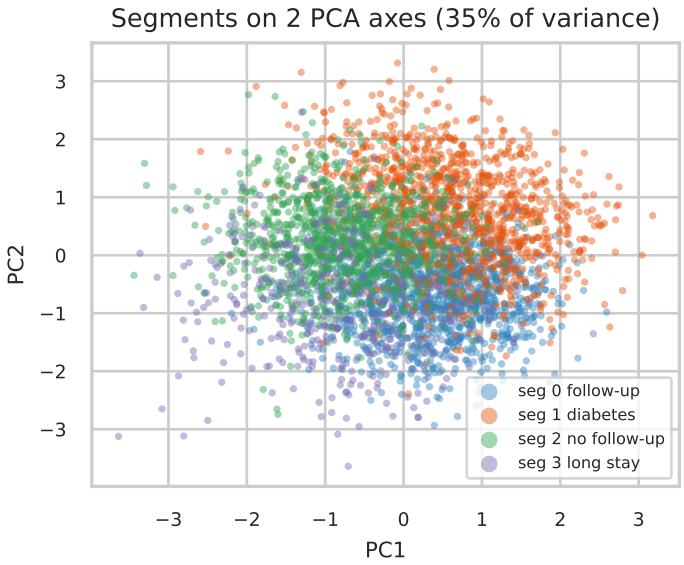
SEG 0 FOLLOW-UP
1286 patients
6.8% readmitted

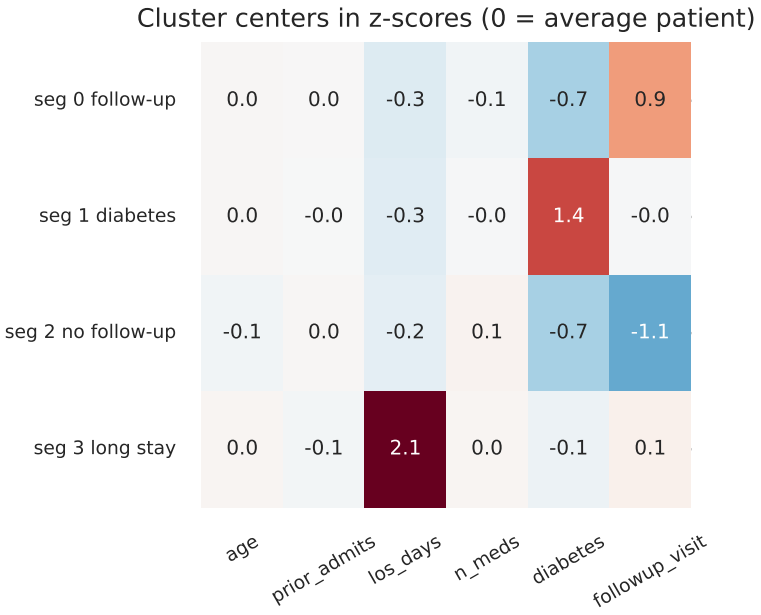
SEG 1 DIABETES
1200 patients
10.1% readmitted

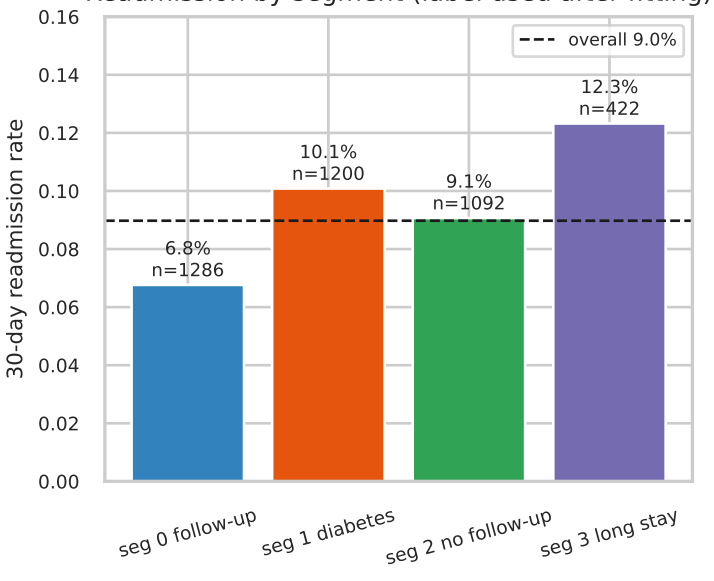
SEG 2 NO FOLLOW-UP
1092 patients
9.1% readmitted

SEG 3 LONG STAY
422 patients
12.3% readmitted









Takeaways
- Scale features or age alone decides the clusters
- Silhouette near 0.2 means the segments overlap; present them as useful groupings, not natural types
- Long-stay patients (segment 3) readmit at 12.3% vs 6.8% for patients who got a follow-up visit
- Clustering is description; to predict risk, use the supervised models from days 14-18
- Tomorrow: dimensionality reduction with PCA, what those two axes really mean

Hands-On Worksheet: ds-genai-90-days / day19

60-90 minutes | notebook: day19/segments.ipynb + day19/README.md

[ ] mkdir day19, conda activate ds90, open day19/segments.ipynb

[ ] Rebuild the days 14-18 data (snippet 1); do NOT use readmit30 to fit

[ ] Look at means and spreads; fit k-means on UNSCALED data (snippet 2)

[ ] Note how unscaled clusters split by age; then scale with StandardScaler

[ ] Loop k = 2..8: record inertia and silhouette (snippet 3)

[ ] Plot the elbow and silhouette charts; write one sentence on k

[ ] Fit k = 4; profile segments in real units (snippet 4)

[ ] Give each segment a plain name a nurse manager would understand

[ ] Bring readmit30 back: rate and size per segment (snippet 5)

[ ] Plot the segments on 2 PCA axes, colored by segment

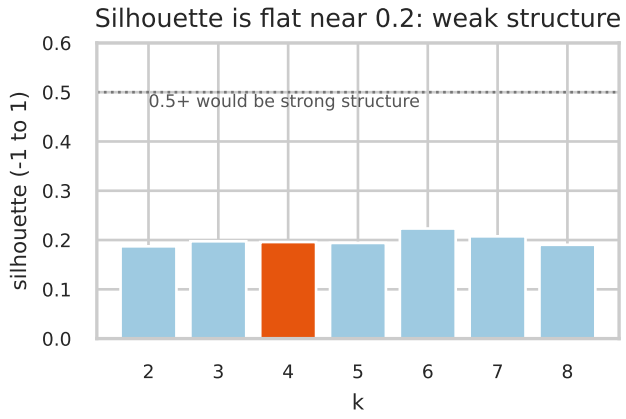
[ ] Stability: refit with 3 other seeds and compare with adjusted Rand index

[ ] README: 3 charts, a segment table, and one action per segment

[ ] git add day19 && git commit && git push (snippet 6)

[ ] Stretch: segment profile table in Snowflake SQL (snippet 7)

```
for k in range(2, 9):
    km = KMeans(n_clusters=k, n_init=10,
                random_state=19).fit(Xs)
    print(k, km.inertia_,
          silhouette_score(Xs, km.labels_))
```



Best silhouette k: ____ Silhouette at k=4: _____ Segment sizes: ____ / ____ / ____ / ____

Readmit rate by segment: ____ / ____ / ____ / ____ Overall: ____ ARI across seeds: ____

Day 19 Self-Check Quiz

1. What makes clustering 'unsupervised'? Which column did we leave out?

_____

2. Describe the two steps k-means repeats until it settles.

_____

3. Why did unscaled k-means split patients only by age?

_____

4. What do inertia and the silhouette score measure? Which way is better?

_____

5. Silhouette is about 0.2 for every k. What does that tell you?

_____

6. Why bring readmit30 back only after fitting the clusters?

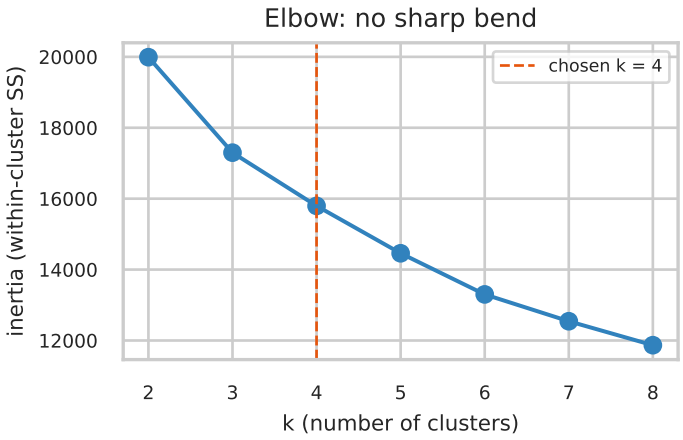
_____

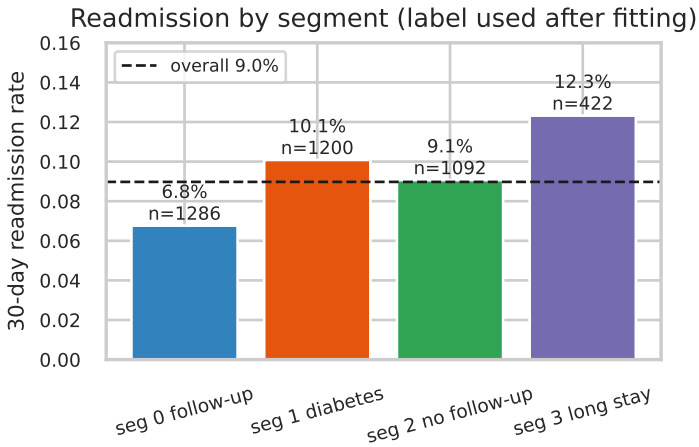
7. What is PCA doing in the scatter plot, and what is it NOT doing?

_____

8. A manager asks for 'the 4 patient types'. How do you present the result?

_____





Answers:
1 no label guides the fit; we cluster on the 6 features only and leave readmit30 out
2 assign every patient to the nearest center, then move each center to the mean of its patients; repeat until stable
3 distance is in raw units, and age (sd ~13) has far more spread than prior admits (sd ~1), so age dominates
4 inertia: squared distance to own center, lower is tighter but always falls with k; silhouette: -1 to 1, higher is better separated
5 the groups overlap a lot; there are no strong natural clusters, so treat segments as useful groupings, not real types
6 so the outcome cannot shape the groups; then the readmission rates are an honest description (6.8% to 12.3%)
7 it squashes 6 scaled columns to 2 axes (35% of variance) only to draw them; the clusters were found in all 6
8 name each by its profile, give size and readmission rate, say segments overlap (silhouette ~0.2) and are stable across seeds